

```
# installs specific versions of the libraries needed to run Phi-3-mini-4k-instruct.
!pip install \
transformers==4.46.3 \
huggingface_hub==0.26.2 \
accelerate==1.1.1 \
tokenizers==0.20.3 \
safetensors
```

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already satisfied: transformers==4.46.3 in /usr/local/lib/python3.12/dist-packages (4.46.3)
already satisfied: huggingface_hub==0.26.2 in /usr/local/lib/python3.12/dist-packages (0.26.2)
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already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (3.29.4)
already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.46.3) (2.0.2)
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already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub==0.26.2) (4.
already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate==1.1.1) (5.9.5)
already satisfied: torch>=1.10.0 in /usr/local/lib/python3.12/dist-packages (from accelerate==1.1.1) (2.11.0+cpu)
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already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1) (1.14.0)
already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1) (3.6.
already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.10.0->accelerate==1.1.1) (3.1.6)
already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.46.3)
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already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.10.0->accele
already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->torch>=1.10.0->accelerate==1.1.
```

```
# It deletes the partially downloaded phi 3 files and starts downloading from strating
!rm -rf ~/.cache/huggingface/hub/models--microsoft--Phi-3-mini-4k-instruct
```

```
# transformer = Hugging Face Python library that provides transformer-based models

#
=== IMPORTING ===

# AutoModelForCausalLM = Automatically selects and loads the suitable causal language model
# AutoTokenizer = Automatically loads the suitable tokenizer for the loaded model
# torch = Transformers uses PyTorch as the backend for running the models

from transformers import AutoModelForCausalLM, AutoTokenizer

# Model name on Hugging Face you want to load
model_name = "microsoft/Phi-3-mini-4k-instruct"

# Automatically loads the pretrained model
model = AutoModelForCausalLM.from_pretrained(
    model_name,

    device_map="auto",          # Loads the model on GPU/cpu which is available (For GPU use = "cuda")

    torch_dtype="auto",        # Automatically selects the datatype for computation

    trust_remote_code=True,    # Allows the custom code to run if needed
)

# Load the tokenizer that automatically converts the text into tokens
# and generated tokens back into text
tokenizer = AutoTokenizer.from_pretrained(model_name)
```

Loading checkpoint shards: 100%

2/2 [00:01<00:00, 1.98it/s]

```
generator=pipeline(
    "text-generation",          # Tells the transformers to generate the text
    model=model,
    tokenizer=tokenizer,        # Uses Previously loaded tokenizer and model
    return_full_text=False,     # Output contains only newly generated tokens not the input ones
    max_new_tokens=500,        # MAX no.of. new tokens to generate
    do_sample=False             # This tells how the next token is generated (Always chooses the highest probability)
)
```

```
# Chat style input It has 2 keys "role" = Who is speaking , content = message send by user

messages=[
    { "role":"user" , "content" : "Create a funny joke about chickens."}
]

# Pipeline internally does Converts chat messages into the model's expected prompt format,Tokenizes the prompt , Runs the model
output=generator(messages)

# Model generates a list of output select the first generated output
print(output[0]["generated_text"])

Why did the chicken join the band? Because it had the drumsticks!
```